

allowed to freeze as the hydrated colloid structure will be irreversibly damaged.

12 Incompatibilities

The point of zero charge is related directly to the Al:P atomic ratio. Therefore, the substitution of additional phosphate groups for hydroxyl groups will lower the point of zero charge. Substitution of carbonate, sulfate, or borate ions for hydroxyl groups will also lower the point of zero charge.

13 Method of Manufacture

Aluminum phosphate adjuvant is formed by the reaction of a solution of aluminum chloride and phosphoric acid with alkali hydroxide.

14 Safety

Aluminum phosphate adjuvant is intended for use in parenteral vaccines and is generally regarded as safe. It may cause mild irritation, dryness, and dermatitis on skin contact. It may also cause redness, conjunctivitis, and short-term mild irritation on eye contact. Ingestion of large amounts of aluminum phosphate adjuvant may cause respiratory irritation with nausea, vomiting, and constipation. Inhalation is unlikely, although the dried product may cause respiratory irritation and cough. Type I hypersensitivity reactions following parenteral administration have also been reported.⁽²⁾

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Eye protection and gloves are recommended.

16 Regulatory Status

GRAS listed. Accepted for use in human and veterinary vaccines in Europe and the USA. The limits for use in human vaccines are 0.85 mg aluminum/dose (FDA) and 1.25 mg aluminum/dose (WHO). There are no established limits for use in veterinary vaccines. Reported in the EPA TSCA Inventory.

17 Related Substances

Aluminum hydroxide adjuvant.

18 Comments

The USP 32 monograph for aluminum phosphate (AlPO₄) gel describes aluminum phosphate, which is used as an antacid, not as a vaccine adjuvant.

19 Specific References

- 1 Shirodkar S *et al.* Aluminum compounds used as adjuvants in vaccines. *Pharm Res* 1990; 7: 1282–1288.
- 2 Goldenthal KL *et al.* Safety evaluation of vaccine adjuvants. *AIDS Res Hum Retroviruses* 1993; 9: S47–S51.

20 General References

- Hem SL, Hogenesch H. Aluminum-containing adjuvants: properties, formulation, and use. Singh M, ed. *Vaccine Adjuvants and Delivery Systems*. New York: Wiley, 2007; 81–114.
- Gupta RK *et al.* Adjuvant properties of aluminum and calcium compounds. Powell MF, Newman MJ, eds. *Vaccine Design*. New York: Plenum, 1995; 229–248.
- Lindblad EB. Aluminum adjuvants – in retrospect and prospect. *Vaccine* 2004; 22: 3658–3668.
- Lindblad EB. Aluminum adjuvants. Stewart-Tull DES, ed. *The Theory and Practical Application of Adjuvants*. New York: Wiley, 1995; 21–35.
- Vogel FR, Hem SL. Immunogenic adjuvants. Plotkin SA *et al.*, ed. *Vaccines*, 5th edn. New York: W.B. Saunders, 2008; 59–71.
- Vogel FR, Powell MF. A compendium of vaccine adjuvants and excipients. Powell MF, Newman MJ, eds. *Vaccine Design*. New York: Plenum, 1995; 142.
- White JL, Hem SL. Characterization of aluminum-containing adjuvants. Brown F *et al.*, ed. *Physico-Chemical Procedures for the Characterization of Vaccines*. IABS Symposia Series: Developments in Biologicals, vol. 103. New York: Karger, 2000; 217–228.

21 Authors

SL Hem, PB Klepak, EB Lindblad.

22 Date of Revision

19 February 2009.



Ammonia Solution

1 Nonproprietary Names

BP: Strong Ammonia Solution

PhEur: Ammonia Solution, Concentrated

USP-NF: Strong Ammonia Solution

2 Synonyms

Ammoniac; ammoniacum; ammoniae solution concentrata; aqua ammonia; concentrated ammonia solution; spirit of hartshorn; stronger ammonia water.

3 Chemical Name and CAS Registry Number

Ammonia [7664-41-7]

4 Empirical Formula and Molecular Weight

NH₃ 17.03

5 Structural Formula

See Section 4.

6 Functional Category

Alkalizing agent.

7 Applications in Pharmaceutical Formulation or Technology

Ammonia solution is typically not used undiluted in pharmaceutical applications. Generally, it is used as a buffering agent or to adjust the pH of solutions. Most commonly, ammonia solution (the

concentrated form) is used to produce more dilute ammonia solutions.

Therapeutically, dilute ammonia solution is used as a reflex stimulant in 'smelling salts', as a rubefacient, and as a counter-irritant to neutralize insect bites or stings.⁽¹⁾

8 Description

Strong ammonia solution occurs as a clear, colorless liquid having an exceedingly pungent, characteristic odor. The PhEur 6.0 states that concentrated ammonia solution contains not less than 25.0% and not more than 30.0% w/w of ammonia (NH₃). The USP32–NF27 states that strong ammonia solution contains not less than 27.0% and not more than 31.0% w/w of ammonia (NH₃).

See also Section 17.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for ammonia solution.

Test	PhEur 6.0	USP32–NF27
Identification	+	+
Characters	+	—
Appearance of solution	+	—
Oxidizable substances	+	+
Pyridine and related substances	≤2 ppm	—
Carbonates	≤60 ppm	—
Chlorides	≤1 ppm	—
Sulfates	≤5 ppm	—
Iron	≤0.25 ppm	—
Heavy metals	≤1 ppm	≤0.0013%
Residue on evaporation	≤20 mg/L	—
Limit of nonvolatile residue	—	≤0.05%
Assay (of NH ₃)	25.0–30.0%	27.0–31.0%

10 Typical Properties

Solubility Miscible with ethanol (95%) and water.

Specific gravity 0.892–0.910

11 Stability and Storage Conditions

On exposure to the air, ammonia solution rapidly loses ammonia. Ammonia solution should be stored in a well-closed container, protected from the air, in a cool, dry place. The storage temperature should not exceed 20°C.

12 Incompatibilities

Ammonia solution reacts vigorously with sulfuric acid or other strong mineral acids and the reaction generates considerable heat; the mixture boils.

13 Method of Manufacture

Ammonia is obtained commercially chiefly by synthesis from its constituent elements, nitrogen and hydrogen, which are combined under high pressure and temperature in the presence of a catalyst. Ammonia solution is produced by dissolving ammonia gas in water.

14 Safety

Ingestion of strong solutions of ammonia is very harmful and causes severe pain in the mouth, throat, and gastrointestinal tract as well as severe local edema with cough, vomiting, and shock. Burns to the esophagus and stomach may result in perforation. Inhalation of the

vapor causes sneezing, coughing, and, in high concentration, pulmonary edema. Asphyxia has been reported. The vapor is irritant to the eyes. Strong solutions are harmful when applied to the conjunctiva and mucous membranes. Topical application of even dilute ammonia solutions, used to treat insect bites, has caused burns, particularly when used with a subsequent dressing.^(2–4)

When used as an excipient, ammonia solution is generally present in a formulation in a highly diluted form.

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Care should be used in handling strong or concentrated ammonia solutions because of the caustic nature of the solution and the irritating properties of its vapor. Before containers are opened, they should be well cooled. The closure should be covered with a cloth or similar material while opening. Ammonia solution should not be tasted and inhalation of the vapor should be avoided. Ammonia solution should be handled in a fume cupboard. Eye protection, gloves, and a respirator are recommended.

16 Regulatory Status

Included in the FDA Inactive Ingredients Database (oral suspensions, topical preparations). Included in nonparenteral medicines licensed in the UK. Included in the Canadian List of Acceptable Non-medicinal Ingredients.

17 Related Substances

Dilute ammonia solution.

Dilute ammonia solution

Synonyms Ammonia water

Specific gravity 0.95–0.96

Comments Several pharmacopeias include monographs for dilute ammonia solution. The JP XV, for example, states that ammonia water contains not less than 9.5% and not more than 10.5% w/v of ammonia (NH₃).

18 Comments

Where 'ammonia solution' is prescribed therapeutically, dilute ammonia solution should be dispensed or supplied.

The EINECS number for ammonia solution is 231-635-3.

19 Specific References

- 1 Frohman IG. Treatment of physalia stings. *J Am Med Assoc* 1996; **197**: 733.
- 2 Beare JD *et al.* Ammonia burns of the eye: an old weapon in new hands. *Br Med J* 1988; **296**: 590.
- 3 Payne MP, Delic JL. Ammonia. *Toxicity Review* 24. London: HMSO, 1991; 1–12.
- 4 Leduc D *et al.* Acute and long term respiratory damage following inhalation of ammonia. *Thorax* 1992; **47**: 755–757.

20 General References

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21 Author

PJ Sheskey.

22 Date of Revision

10 January 2009.



Ammonium Alginate

A

1 Nonproprietary Names

None adopted.

2 Synonyms

Alginic acid, ammonium salt; ammonium polymannuronate; E404; *Keltose*.

3 Chemical Name and CAS Registry Number

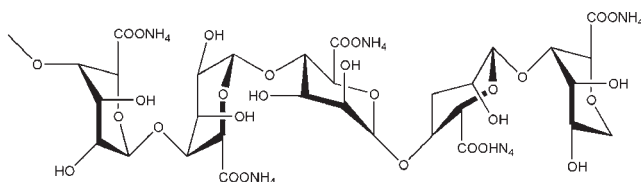
Ammonium alginate [9005-34-9]

4 Empirical Formula and Molecular Weight

(C₆H₁₁NO₆)_n 193.16 (calculated)
217 (actual, average)

Ammonium alginate is the ammonium salt of alginic acid.

5 Structural Formula



The number and sequence of the mannuronate and glucuronate residues shown above vary in the naturally occurring alginate. The associated water molecules are not shown.

6 Functional Category

Diluent; emulsifying agent; film-forming agent; humectant; stabilizing agent; thickening agent.

7 Applications in Pharmaceutical Formulation or Technology

Ammonium alginate is widely used in foods as a stabilizer, thickener and emulsifier. It is also used in pharmaceutical preparations as a color-diluent, emulsifier, film-former, and humectant.

8 Description

Ammonium alginate occurs as white to yellowish brown filamentous, grainy, granular, or powdered forms.

9 Pharmacopeial Specifications

See Section 18.

10 Typical Properties

Moisture content Not more than 15% at 105°C for 4 hours.

Solubility Dissolves slowly in water to form a viscous solution; insoluble in ethanol and in ether.

11 Stability and Storage Conditions

Ammonium alginate is a hygroscopic material, although it is stable if stored at low relative humidities and cool temperatures.

12 Incompatibilities

Incompatible with oxidizing agents and strong acids and alkalis.

13 Method of Manufacture

—

14 Safety

Ammonium alginate is widely used in cosmetics and food products, and also in pharmaceutical formulations such as tablets. It is generally regarded as a nontoxic and nonirritant material, although excessive oral consumption may be harmful.

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled. Eye protection, gloves, and a dust respirator are recommended.

16 Regulatory Status

GRAS listed. Accepted in Europe for use as a food additive. Included in the FDA Inactive Ingredients Database (oral, tablets).

17 Related Substances

Alginic acid; calcium alginate; potassium alginate; propylene glycol alginate; sodium alginate.

18 Comments

Alginates are commonly used in wound dressings.⁽¹⁾ Chitosan and alginates have been used together to produce sponges for use as wound dressings, or matrices for tissue engineering.⁽²⁾ Alginate microspheres have been produced by internal gelation using emulsification methods.⁽³⁾

Although not included in any pharmacopeias, a specification for ammonium alginate is contained in the Food Chemicals Codex (FCC), see Table I.

Table I: FCC specification for ammonium alginate.⁽⁴⁾

Test	FCC 6 ⁽⁴⁾
Identification	+
Arsenic	≤ 3 mg/kg
Ash	≤ 7.0% after drying
Lead	≤ 5 mg/kg
Loss on drying	≤ 15.0%
Assay	18.0–21.0% of CO ₂ , corresponding to 88.7–103.6% ammonium alginate

19 Specific References

- 1 Morgan D. Wounds—what should a dressing formulary include? *Hosp Pharm* 2002; 9(9): 261–266.
- 2 Lai HL *et al*. The preparation and characterization of drug-loaded alginate and chitosan sponges. *Int J Pharm* 2003; 251(1–2): 175–181.
- 3 Chan LW *et al*. Production of alginate microspheres by internal gelation using an emulsification method. *Int J Pharm* 2002; 242(1–2): 259–262.
- 4 *Food Chemicals Codex*, 6th edn. Bethesda, MD: United States Pharmacopeia, 2008; 44.

20 General References

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21 Authors

SA Shah, D Thassu.

22 Date of Revision

30 June 2008.